RAG-Enhanced Evidence Recommendation in Financial Legal Resolutions

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Abstract

The complexity of legal documents, particularly in financial legal disputes, poses significant challenges for both experts and automated systems. This study introduces a system leveraging Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) technology to recommend key evidence in financial advisor dispute cases. Unlike traditional legal AI tasks focused on outcome prediction, our approach emphasizes supporting judicial reasoning by recommending key evidence relevant to adjudication. We constructed a dataset of 371 annotated cases from Taiwan, spanning 25 years, including claims, judicial opinions, and final judgments, with annotations highlighting key evidence. Using RAG, our system retrieves and generates evidence recommendations grounded in analogous past cases while maintaining temporal and contextual consistency. This methodology enhances judicial efficiency and supports equitable legal decision-making by streamlining the recommendation of critical evidence.

CCS Concepts

• **Computing methodologies → Natural language processing**.

Keywords

Evidence Recommendation, Retrieval Augmented Generation, Financial Advisory

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1 Introduction

In recent years, the adoption of Natural Legal Language Processing (NLLP) has expanded significantly, with applications in areas such as legal question answering [5, 19], court ruling prediction [8, 14], legal text retrieval and generation [2, 7], and legal text summarization [3, 15]. These advancements not only enhance the efficiency of legal processes but also contribute to greater transparency and accessibility in legal services. Despite these strides, the complexity and variability of legal documents continue to pose challenges, even for experts. This is particularly evident in niche domains like financial legal disputes, where specialized knowledge is often required to interpret evidence and assess claims.

Financial legal disputes, especially those involving financial advisors and their clients, have become increasingly common due to issues like information asymmetry, misleading sales practices, and product mismatches. Advisors may exploit clients' lack of expertise to promote unsuitable products, or clients may resist accepting losses attributable to market risks or advisory missteps. These disputes often culminate in court proceedings where the outcome hinges on the presence and interpretation of key evidence. For judges, the task of sifting through voluminous legal documents to recommend the most relevant evidence is labor-intensive and time-consuming, underlining the need for automated tools to assist in evidence recommendation.

To address this challenge, we propose a system that leverages Large Language Models (LLMs) to assist in evidence recommendation for financial legal disputes. Unlike conventional legal AI tasks that focus on predicting case outcomes, our approach emphasizes the judgment process, particularly recommending the evidence that could influence judicial decisions. By recommending evidence, we aim to assist both plaintiffs and defendants in preparing comprehensive legal documentation and to streamline the judicial process.

We discuss the role of Retrieval-Augmented Generation (RAG) technology in the proposed task. RAG combines retrieval and generation mechanisms, enabling models to first retrieve relevant information from a database and then use it as context for generating accurate and contextually rich outputs. In the legal domain, where precedents and historical cases are vital, RAG enhances the model's ability to generate evidence recommendations grounded in similar past judgments.

Recognizing the scarcity of Chinese civil legal datasets tailored to this task, we have manually compiled a dataset comprising financial advisor dispute cases in Taiwan from the past 25 years. This dataset includes party assertions, judicial opinions, final judgments, and compensation amounts. Additionally, we annotated key evidence referenced in judicial opinions to create a foundation for training and evaluating our system. By combining these annotated cases with RAG technology, we aim to improve the accuracy and relevance of evidence recommendations.

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To sum up, the contributions of our work are as follows:

(1) **Introduction of a Novel Task**: We introduce a new task within legal AI—the recommendation of key evidence from legal judgments. This task shifts the focus from predicting outcomes to supporting the judgment process, aiding stakeholders in recommending critical evidence for their cases.
(2) **Creation of a Specialized Dataset**: We developed a dataset, FinDispute,[1] encompassing all financial advisor dispute cases in Taiwan over the past 25 years. This dataset includes claims from both parties, judicial opinions, final judgments, and compensation details. It is uniquely annotated to highlight key evidence identified by judges, providing a valuable resource for research and development in legal AI.
(3) **Application of RAG in Legal AI**: We applied Retrieval-Augmented Generation (RAG) technology to this novel task. By leveraging claims and key evidence from past cases, RAG supports more precise and contextually relevant evidence recommendations.

2 Related Work

Natural Language Processing (NLP) has been widely applied to the field of legal text processing, addressing the inherent challenges posed by the unstructured and complex nature of legal documents. Early approaches primarily relied on rule-based systems and traditional machine learning models. For instance, Bartolini et al. [1] established a framework for manual information extraction from legal documents, while Vacek and Schilder [16] utilized Conditional Random Fields to identify case outcomes. These methods demonstrated utility in extracting structured information but were often constrained by scalability and adaptability to diverse legal contexts.

Recent studies have also explored hybrid approaches that combine deep learning with other techniques. For example, Lippi et al. [13] integrated deep learning and Support Vector Machines (SVMs) to identify unfair clauses in online service agreements, framing it as a sentence classification task. Xia et al. [18] utilized Word2Vec embeddings with legal corpora to improve document similarity analysis, while Nanda et al. [11] developed unsupervised lexical and semantic similarity models for analyzing European legal corpora. These advancements have significantly broadened the scope of NLP applications in the legal domain, including citation ranking [10] and clause detection.

RAG represents a novel approach in NLP, combining retrieval mechanisms with generative models to produce contextually enriched outputs. First introduced by Lewis et al. [6], RAG demonstrated improved performance across various question-answering tasks by incorporating external knowledge bases into the generative process. Mao et al. [9] further applied RAG to open-domain extractive question answering, achieving state-of-the-art results by integrating RAG with the BM25 retrieval model. In the legal domain, RAG has shown promise for tasks requiring high contextual fidelity. Ram et al. [12] proposed a method for splitting and vectorizing text chunks to calculate semantic similarity during retrieval. Wiratunga et al. [17] applied RAG to legal question answering, utilizing past cases to enhance the accuracy of generated responses. Hou et al. [4] extended RAG to citation recommendation, leveraging retrieved citations to improve the quality of legal analyses.

Despite its potential, existing RAG implementations often neglect temporal relevance, which is crucial in legal contexts. To address this, our approach incorporates a chronological constraint, ensuring that recommendations only reference judgments made prior to the current case. By structuring the retrieval process in this manner, our system aligns with the legal principles of precedence and temporal consistency, enhancing its utility in evidence recommendation tasks.

While previous works have demonstrated the applicability of NLP and RAG to legal tasks, they often focus on tasks like classification, summarization, or general question answering. Our work differs by introducing evidence recommendation as a distinct task, emphasizing the judgment process rather than outcome prediction. Additionally, the creation of a specialized dataset for financial advisor disputes in Taiwan and the chronological integration of RAG address gaps in existing research.

3 Dataset and Task Definition

The task of evidence recommendation, as defined in this work, focuses on generating relevant and critical pieces of evidence that may be instrumental in adjudicating financial legal disputes. This approach aims to assist judges in their pre-judgment process by proposing evidence based on the claims provided by both the plaintiff and the defendant. Unlike evidence extraction, which involves identifying specific excerpts from documents, evidence recommendation emphasizes the identification of key information that could directly support the resolution of legal disputes.

To support this task, we constructed a specialized dataset through a meticulous data collection and annotation process. This dataset comprises financial legal cases collected from the Judicial Yuan's judgment system in Taiwan,[2] covering disputes involving financial advisors over the past twenty-five years. The cases were carefully filtered using targeted keywords, ensuring relevance to the domain of financial advisory disputes, while excluding unrelated judgments to maintain dataset precision. Each case document includes a comprehensive representation of the dispute, encompassing the plaintiff's and defendant's claims, judicial opinions, and final verdicts.

The resulting dataset consists of 371 annotated cases, of which 93 resulted in guilty verdicts and 278 were adjudicated as not guilty. By focusing on cases involving financial advisor disputes, this dataset fills a critical gap in the legal domain. The inclusion of annotated key evidence ensures that the dataset is particularly suited for evidence recommendation tasks, offering rich insights into how critical information influences judicial outcomes in financial disputes.

4 Methodology

RAG is a framework designed to enhance the capabilities of language models by incorporating external knowledge retrieval into the text generation process. When a query is received, RAG employs an information retrieval component, such as dense vector indexing or similar retrieval algorithms, to fetch relevant documents from external knowledge sources like databases or repositories. These

[1]https://github.com/NYCU-NLP-Lab/FinDispute

[2]https://www.judicial.gov.tw/en/mp-2.html

documents provide critical contextual information or supporting materials relevant to the input query.

In applying RAG to financial legal disputes, we developed a methodology tailored to the specific needs of the legal domain. The process begins by dividing the dataset, which includes claims and their corresponding key evidence, into multiple batches. For the first batch, the claims are input into an LLM, which generates the corresponding key evidence. This evidence is then added to the texts in the batch to create updated examples. These updated examples are vectorized and stored in a database, establishing a foundation for subsequent iterations.

As additional batches are processed, the claims from the current batch are vectorized and compared against the stored examples in the database to identify the top k most similar texts. These similar texts are incorporated into the context of the current query, enriching the input provided to the model. The model then generates answers that leverage both the newly retrieved information and the previously stored examples. This iterative process ensures that the generated responses become progressively more accurate and informed as the database expands.

A critical feature of our methodology is the incorporation of a temporally aware retrieval mechanism. In the legal domain, judicial decision-making often depends on referencing precedents and analyzing past cases. Our system ensures that only rulings and information available up to the time of the current case are considered. By excluding future or unrelated data, we maintain the temporal validity and legal consistency of the outputs. This is especially important because legal statutes, interpretations, and precedents evolve over time, and adherence to these temporal constraints is crucial for generating reliable recommendations.

This approach aligns with the referential nature of legal reasoning, enhancing the reliability and contextual accuracy of the model's outputs. By systematically referencing relevant precedents, our methodology emulates the structured reasoning process employed in judicial systems, thereby improving the model's ability to generate legally grounded and contextually appropriate answers.

Inspired by previous works, we adopt BM25 [9] and language model vectors [12] for retrieving historical cases.

5 Experiment

5.1 Experimental Setup

In this experiment, we used the GPT-3.5-turbo-16k model to address the issue of large document lengths in judgments. To ensure the reproducibility of results, we set the temperature to 0. The model's context window size is 16,385 tokens, and the batch size is 10, meaning that 10 cases were input into the model at a time. After obtaining the response, the correct answers were appended and added to the database. During retrieval, we selected the top k most similar texts to the current input, with k set to 5 in our experiment.

We conducted an experiment where, initially, no prompts were provided. Instead, we used the claims from both parties as input and asked the model to generate the corresponding key evidence. This approach served as the baseline for comparison. Subsequently, we implemented the RAG method and compared its performance with the baseline to evaluate the effectiveness of RAG in improving the accuracy and relevance of the generated outputs.

Table 1: Experimental results.

IR Method	Precision	Recall	F1
-	0.281	0.622	0.387
Random	0.338	0.635	0.441
BM25	0.340	0.640	0.444
Vector-Based	**0.354**	**0.672**	**0.464**

To further analyze the impact of the retrieval component, we designed a control experiment. In this experiment, five texts were randomly selected as templates and included in the prompts to observe how these randomly selected templates affected the model's generation of key evidence. By comparing these three different experimental setups (i.e., no prompts, RAG method, and random template selection), we were able to comprehensively assess the contribution of the retrieval-augmented generation method to enhancing the model's generative performance. Precision, recall, and F1 score are adopted for evaluation.

5.2 Experimental Result

Table 1 presents the experimental results. Incorporating retrieved prior instances for the LLM to reference, as opposed to merely prompting the LLM to recommend key evidence, consistently demonstrates improved performance across all metrics. These results demonstrate the importance of RAG strategies for key evidence recommendation tasks.

When no retrieval method is used (denoted as "-"), the results highlight a fundamental limitation in the LLM's standalone capacity to identify key evidence, with precision at 0.281, recall at 0.622, and an F1 score of 0.387. These metrics suggest that while the LLM is capable of recalling a broader range of evidence, it often lacks the precision needed to filter relevant instances effectively. Introducing random retrieval yields noticeable improvements, with the F1 score rising to 0.441. This demonstrates that even when references are chosen arbitrarily, they provide contextual grounding that aids the LLM in refining its outputs. The BM25 retrieval method, which uses term frequency and inverse document frequency to rank documents, further improves the performance. Among all methods examined, the vector-based retrieval approach achieves the best performance. This superior performance can be attributed to the semantic matching capabilities of vector-based models, which effectively capture conceptual relationships between the input and retrieved cases. The increase in both precision and recall indicates that vector-based retrieval not only identifies relevant instances but does so with minimal noise, leading to a more balanced and robust performance.

These findings reveal the critical role of retrieval relevance in improving key evidence recommendation tasks. Regardless of the IR method employed, incorporating retrieved instances consistently enhances the LLM's effectiveness. The results particularly emphasize the potential of semantic retrieval techniques, such as vector-based methods, to achieve deeper contextual understanding. Consequently, the use of advanced retrieval techniques in RAG frameworks holds significant promise for further advancing the performance of LLMs in complex recommendation tasks.

5.3 Analysis of Evidence Types and Judicial Outcomes

We further examine the types of evidence that may influence judicial decisions. To gain a deeper understanding of the factors driving these outcomes, we analyze the Pointwise Mutual Information (PMI) values of keywords associated with consumer victories and losses.

In cases where customers achieve favorable outcomes, keywords such as "Fraud," "Default," "Safe," "Transfer," "Forgery," and "Embezzlement" exhibit a highly positive PMI value of 18.356. These terms are closely associated with misconduct, high-risk behaviors, or legally significant actions, suggesting their critical role in supporting customers' claims during financial disputes. The strong positive correlation of these keywords with successful outcomes highlights the importance of uncovering and emphasizing evidence of fraudulent or unethical practices in the resolution process.

Conversely, for cases where customer complaints are unsuccessful, keywords such as "Internal control," "Deficiency," "Application," "Misrepresentation," and "Issuance" demonstrate negative PMI values. These terms point to procedural or technical issues, such as weaknesses in internal controls, inaccuracies in applications, or misrepresentations, which often result in adverse outcomes for customers. The negative PMI values underline the frequent association of these operational deficiencies with unsuccessful complaints.

The stark contrast between these two sets of keywords highlights the dichotomy in the nature of factors influencing financial dispute outcomes. Keywords associated with successful complaints are more likely to signify egregious behavior or systemic violations, whereas unsuccessful complaints tend to revolve around operational inefficiencies or procedural lapses. This differentiation underscores the significance of robust evidence and precise documentation in shaping outcomes.

6 Conclusion

In this paper, we create the specialized FinDispute dataset, which compiles 25 years of financial advisor dispute cases from Taiwan, annotated with key judicial evidence. This dataset provides a foundational resource for advancing research in legal AI. Additionally, our methodology demonstrates the effectiveness of RAG in enhancing the accuracy and relevance of evidence recommendations. By analyzing evidence types and judicial outcomes, we uncover the pivotal role of keywords in shaping case resolutions, highlighting the importance of evidence that aligns with factors like fraud or misconduct. These insights not only guide the development of automated systems but also assist stakeholders in presenting and interpreting critical evidence effectively.

Future work could focus on expanding the dataset to cover a broader range of legal domains and jurisdictions, as well as refining retrieval algorithms to incorporate evolving legal precedents.

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